

Non-deterministic Computation

collapse-bind

Description: Evaluates a MeTTa operation and returns an expression containing all alternative evaluations in the form `(Atom Bindings)`.

Parameters:

- Atom: The MeTTa operation to be evaluated.

Return: An expression of alternative evaluations with bindings.

Example:

```
(= (bin) 0)
(= (bin) 1)
!(collapse-bind (bin)) ; Returns (0 { }), (1 { }) nondeterministically
```

superpose-bind

Description: Takes the result of `collapse-bind` and returns only the result atoms, discarding the bindings.

Parameters:

- Expression: An expression in the form `(Atom Bindings)`.

Return: A non-deterministic list of atoms.

Example:

```
!(superpose-bind ((A (Grounded ...)) (B (Grounded ...)))) ; returns the equivalent of (superpose (A B))
```

superpose

Description: Turns a tuple into a nondeterministic result.

Parameters:

- Tuple: Tuple to be converted.

Return: Argument converted to nondeterministic result

Example:

```
!(superpose (A B C)) ; returns A, B, C nondeterministically
```

collapse

Description: Converts a nondeterministic result into a tuple.

Parameters:

- Atom: Atom which will be evaluated.

Return: Tuple

Example:

```
!(collapse (superpose (A B C))) ; returns (A B C)
```